

Not easy to “like”: How does cognitive load influence user engagement in online reviews?

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ABSTRACT

User engagement (e.g., likes, shares, and comments) is widely recognized as critical to business success. Although existing studies have explored the determinants of user engagement, relatively little attention has been paid to cognitive load. This study, based on cognitive load theory, expectation confirmation theory, and the stressor-strain-outcome framework, examines the heterogeneous effects of intrinsic, extraneous, and germane cognitive load on user engagement in Study 1, the moderating effects of intrinsic, extraneous, and germane cognitive load variance (i.e., described as cognitive load changes induced by processing from one review to another review) in Study 1, and their underlying mechanisms in Study 2. Results indicate that: (1) intrinsic and extraneous cognitive load negatively influence user engagement, while germane cognitive load has a positive effect, (2) intrinsic/extraneous cognitive load variance accentuates the negative effect of intrinsic/extraneous cognitive load, while germane cognitive load variance strengthens the positive effect of germane cognitive load, and (3) perceived fatigue negatively mediates the effects of intrinsic and extraneous cognitive load but positively mediates the effect of germane cognitive load. Our findings contribute to the literature on user engagement and offer practical insights for optimizing online review systems.

1. Introduction

User engagement, described as likes, comments, shares, and click-throughs—with the given information [33], has been recognized as important to business success [58,59], such as promoting product sales [5] and improving firm performance [70]. A survey conducted by Business of Apps indicates that 90 % of users who engage with an app at least once a week have a high possibility of becoming long-term users.¹ The positive effect of user engagement is also salient in the context of online reviews. To help consumers efficiently process review information, review platforms encourage consumers to vote for informative reviews [58], which provides a basis for platforms to optimize review rankings. As a pre-step to leverage the significant role of user engagement in the context of online reviews, it is essential to understand what factors influence user engagement.

Previous research has discussed the antecedents of user engagement through the lens of content- and creator-based feature perspectives [25,64]. The content-based feature perspective emphasizes how the characteristics of novel information affect user engagement. Wang, Ngai

and Li [58], for example, demonstrated that review informativeness positively influences consumer voting behavior. The creator-based feature perspective describes how the characteristics of information providers influence user engagement. Kuan, Hui, Prasarnphanich and Lai [32], for example, denoted that reviewer credibility has a high possibility of eliciting consumer voting behavior in the context of online reviews. However, limited studies are advanced from the cognitive load perspective. In online review contexts, cognitive load—that is, described as the load imposed on an individual's cognitive system when performing a given task, and which is usually associated with the limitations of the human working memory [39]—emerges as a significant concern [58,63]. Understanding the effect of cognitive load, especially different types of cognitive load, is essential for the following two reasons.

Theoretically, although a few studies have discussed cognitive load, the effects of different types of cognitive load are scarcely discussed. In many cases, cognitive load has been oversimplified as an overall item, as much evidence supports the claim that cognitive load acts as an inhibitor, harming individuals in digesting novel information [2,24,58].

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¹ J. Wiafe, User Engagement: How to Drive Growth with User Acquisition and Retention, Yodel Mobile, (2024).

However, the effect of cognitive load may vary depending on the type of load, that is, some types of cognitive load may lead to undesirable outcomes, while others can have positive effects. Differentiating cognitive load types is beneficial for forum operators in understanding the specific impact of cognitive load and developing precise strategies to manage cognitive load. Cognitive load theory provides a basis, claiming there are three distinct types of cognitive load—*intrinsic*, *extraneous*, and *germane* cognitive load [24,39], which provides a solid basis to deepen the understanding of the different effects of cognitive load. *Intrinsic* cognitive load refers to the information complexity or the interaction between information elements [42], such as task complexity [13]. *Extraneous* cognitive load is related to the presentation and organization of information [40], such as idea presentation [13]. *Germane* cognitive load reflects processors' effort to comprehend and memorize novel information, facilitating schema construction [37], such as procedural guidance [13].

Practically, review forums are criticized for information overload [63]. Filtering out targeted information confuses and overloads consumers [63]. The overwhelming information may harm information digestion by exhausting cognitive resources, which inhibits user engagement [63]. Additionally, cognitive load may cause individuals to conduct irrational behavior. Alashoor, Keil, Smith and McConnell [2], for example, indicated that when cognitive depletion is high, the inhibiting effect of privacy concerns on disclosure behavior is weak. Based on the in-depth understanding of the heterogeneous influences of *intrinsic*, *extraneous*, and *germane* cognitive load, platform practitioners can take specific actions to decrease (increase) *intrinsic* and *extraneous* (germane) cognitive load in the information provision process, rather than be guided by the conclusion that all cognitive load is destroyed. Based on the above illustration, we ask the first research question: *How does intrinsic, extraneous, and germane cognitive load heterogeneously influence user engagement?*

Furthermore, it is also important to understand how the effects of different cognitive load types are moderated by situational factors, described as the association between individual information in a review set [59], such as sentiment dispersion [59] and review dissonance [66]. In this study, we use *intrinsic*, *extraneous*, and *germane* cognitive load variance—which is described as the corresponding cognitive load change from processing one to another information—to depict situational factors. Although previous studies have conducted related discussions on the effect of cognitive load [11,24,54], how this effect changes with cognitive load variance has drawn limited attention. In practice, when users search for available information, one piece of information may not be enough to support users in making decisions [59]. As information processing transitions from one piece of information to another, the corresponding cognitive load is expected to change accordingly. Understanding cognitive load variance can help platforms rank reviews. Hence, we ask the second research question: *How does intrinsic/extraneous/germane cognitive load variance moderate the effect of intrinsic/extraneous/germane cognitive load on user engagement?*

Besides, to deepen the understanding of the influence of cognitive load, it is of paramount importance to unbox its underlying mechanism. The stressor-strain-outcome framework, describing how users react to external stressors, provides us with a theoretical basis to discuss its underlying mechanism [31,71]. This study contextualized *strain* as perceived fatigue. Perceived fatigue is important for the following reasons. First, the undesirable outcomes of perceived fatigue have been well recognized [4,17,48]. A deep understanding of perceived fatigue could guide platform operators to take targeted actions to alleviate its undesirable outcomes. Second, perceived fatigue is frequently used to depict *strain* [4,71]. Using perceived fatigue broadens the stressor-strain-outcome framework to the cognitive load context. Hence, we ask the third research question: *How does perceived fatigue mediate the effects of intrinsic, extraneous, and germane cognitive load on user engagement?*

To answer research questions, we conducted an empirical analysis using a data set collected from JD.com and two online experiments. In

Study 1, based on the cognitive load theory and expectation confirmation theory, we discuss the heterogeneous effects of *intrinsic*, *extraneous*, and *germane* cognitive load on user engagement, and how these effects change with corresponding cognitive load variance (Hypotheses 1–6) using negative binomial regression. In Study 2, we mainly focus on discussing the mediating effect of perceived fatigue (Hypotheses 7–9) using PROCESS MODEL 4.

The present study provides several contributions to the theory. First, to the best of our knowledge, our study is among the first to discuss the influences of the three types of cognitive load on user engagement, enriching user engagement literature from its antecedent perspective. Second, we examine how the relationships between the three types of cognitive load and user engagement change with corresponding cognitive load variance, which contributes to the cognitive load and user engagement literature by going beyond examining single information and focusing on the level of information sets. Third, this study also unboxes the underlying mechanisms of the effects of the three types of cognitive load, contributing to the cognitive load literature by highlighting its effect pathways.

2. Literature review

2.1. Stressor: Cognitive load

Cognitive load refers to the load imposed on an individual's cognitive system when performing a given task and is usually associated with the limitations of working memory [39]. With the proliferation of available information, cognitive load has become an important issue for information systems research [2,53,71]. However, most studies consider cognitive load as an overall item, ignoring the heterogeneous effects of its different types. As research has progressed, different types of cognitive load have attracted considerable attention among academics. For example, Hu, Hu and Fang [24], based on cognitive load theory, categorized cognitive load into *intrinsic*, *extraneous*, and *germane* cognitive load. Their study denoted that the cognitive load induced by searching for targeted information on a website mediates the effect of the site's navigation structure on user satisfaction [24]. Wang, Yang, Liu, Cao and Ma [56] also divided cognitive load into *intrinsic*, *extraneous*, and *germane* cognitive load, demonstrating that task complexity significantly moderates the relationship between website complexity and users' visual attention and behavior (e.g., fixation count and task completion time). While these studies have applied cognitive load theory, they have not fully explored how different types of cognitive load produce distinct effects.

As an important step forward, Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13], based on cognitive load theory, attempted to deepen the understanding of cognitive load by examining the relationships between different types of cognitive load (i.e., *intrinsic*, *extraneous*, and *germane* cognitive load) and task performance (i.e., idea convergence quality). Like their research, our study decomposes cognitive load into *intrinsic*, *extraneous*, and *germane* load, but differs in that we examine both the association between cognitive load and user engagement and its underlying mechanisms, which are ignored by previous studies.

2.2. Strain: Perceived fatigue

Perceived fatigue is described as a subjective user experience containing negative feelings, including tiredness, annoyance, anger, and disappointment [44]. Load-related factors, such as social media overload [4] and information overload [71], are responsible for this psychological reaction. For example, Baj-Rogowska [4] demonstrated that time cost increases social media fatigue, which leads to discontinuance of usage. Cram, Proudfoot and D'Arcy [17] denoted that the quantity of security policy communications and activities is a salient determinant of security fatigue. Zhang, Zhao, Lu and Yang [71] documented that system

feature overload, information overload, and social overload can increase users' perceptions of social network fatigue, which leads to discontinuous usage intention. However, most existing studies have been conducted in social media or security contexts [4,71], lacking discussion of the fatigue induced by cognitive load in online review settings. Perceived fatigue is significant in our research context because it can act as a psychological reaction to cognitive load induced by overwhelming information [2]. In the context of online reviews, perceived fatigue plays a crucial role as users encounter lots of online reviews. This information abundance can deplete cognitive resources, potentially triggering fatigue responses.

2.3. Outcome: User engagement

User engagement is described as behaviors such as likes, comments, shares, and click-throughs in response to given information [33]. Given the potential of user engagement in business [70], such as filtering out informative reviews [58] and expanding the reach of information [59], its antecedents have drawn considerable academic attention [25,64]. Existing studies have considered information content as a salient driver [19,64]. For example, Yang, Ren and Adomavicius [64] demonstrated that user engagement is not only determined by the valence of a post but also by the specific ways in which a post is positive or negative. Feng, Hui, Deng and Jiang [19] indicated that positive emotional words positively influence the forwarding of government microblog posts because followers want to show a positive outlook for others. Recently, scholars have paid attention to differentiating text- and image-based content [55]. For firm-generated content, Gu, Chan, Kwon, Dhaoui and Septianto [21] demonstrated that informational cues are more effective in influencing user engagement when these cues are presented in visual content (vs. presented in textual content). For user-generated content, Zhou, Xiu, Wang and Yu [72] denoted that the combination of text and images can induce more information dissemination than plain text.

Besides, the existing studies also consider the effect of creator-based characteristics [30,61]. For example, Kim, Jiang, Duhachek, Lee and Garvey [30] demonstrated that consumers have a higher possibility to disclose personal information (a type of user engagement) to an AI than a human agent. Choi [15] documented that the effect of users' verified purchase badges on online voting behavior (a type of user engagement) is less significant for extreme online reviews (than for moderate reviews). However, the existing studies scarcely discuss the antecedents of user engagement from the different types of cognitive load perspective [16,21,23,30]. To bridge this research gap, our study enters the discussion by discussing the effects of the three types of cognitive load.

3. Theoretical background and hypothesis development

3.1. Theoretical background

3.1.1. Stressor-strain-outcome framework

The stressor-strain-outcome framework [31] emphasizes the process of how users react to environmental stressor stimuli. The framework includes three conceptual blocks: *stressor*, *strain*, and *outcome* [71]. These three blocks are sorted in sequence. The stressors evoke users' strain, and then users respond to the elicited strain. *Stressor* describes the things, events, stimuli, demands, and conditions individuals face and encounter in daily life as factors generating stress to individuals [10]. *Strain* is described as "the behavioral and psychological outcomes of stress that are observed in individuals" [43]. *Outcome* is the intentions and behaviors of individuals determined by individual strain elicited by stressors [71]. The stressor-strain-outcome framework is widely employed to discuss how the strain induced by stressors influences individuals' reactions, especially in the context of overwhelming stimuli [48,71]. For example, Zhang, Zhao, Lu and Yang [71] demonstrated that information overload (depicted as a stressor) increases users'

perceptions of social network fatigue (depicted as strain), which in turn increases users' dissatisfaction with social network sites (depicted as outcome). This theory fits the research on object overload, such as information overload [71] and social media use intensity [57].

Related to the current study, the stressor-strain-outcome framework can be used to provide a basis to understand the effects of the three types of cognitive load (depicted as a stressor) on user engagement (depicted as outcome) through perceived fatigue (depicted as strain) for the following two reasons. On the one hand, the stressor-strain-outcome framework fits the focus—that is, cognitive load—of this study. When individuals process novel information, they need to allocate cognitive resources. Because of the limited cognitive capacity [41], cognitive load depletion induced by information processing can be considered as a corresponding stressor. With the increment in available information, users need to allocate extra cognitive resources to digest novel information, which can be considered a burden on users. On the other hand, this framework can provide us with a framework to not only discuss the effects of three types of cognitive load but also corresponding effect pathways (depicted as a strain). As claimed in the stressor-strain-outcome framework [71], when users face external stressors, such as information overload [71] and technology-driven stressors [14], their internal strain is evoked, such as anxiety [14] and fatigue [4], which further induces behavioral and psychological reactions. This following logic provides our study with a basis to understand how different types of cognitive load influence users' perceptions of fatigue, which further influences user engagement.

3.1.2. Cognitive load theory

Cognitive load theory explains the mechanism of how individuals cognitively react to a given task [13]. According to cognitive load theory, working memory (also named short-term memory) and long-term memory make up the cognitive structure [50]. Because the ability of working memory to process novel information is limited [13], overwhelming information may induce undesirable effects when the number of information exceeds the premium quantity [18]. As claimed in the cognitive load theory, three types of cognitive load are included: intrinsic, extraneous, and germane cognitive load.

Specifically, intrinsic cognitive load stems from the information complexity or the interaction between information elements [42]. Seufert, Jänen and Brünken [47] indicated that the complexity of learning tasks increases intrinsic cognitive load. Extraneous cognitive load is related to the presentation and organization of information [40]. Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13], for example, manipulated extraneous cognitive load by the provision of organization function [13]. In the high extraneous cognitive load condition, information is listed irregularly. By contrast, information is categorized by types in the low extraneous load condition [13]. Germane cognitive load reflects processors' effort to comprehend and memorize novel information, facilitating schema construction [13]. With the increase in germane cognitive load, the available cognitive resources to digest newly acquired information are spaced [50]. Sweller, van Merriënboer and Paas [51] demonstrated that the germane cognitive load induced by instructional design is beneficial for learning because the design can guide user attention to schema acquisition by directing users' attention towards relevant learning processes. Cognitive load theory fits our research for the following reasons. On the one hand, cognitive load theory provides a theoretical basis for our research to examine the influences of three types of cognitive load. On the other hand, this theory rationalizes our discussion on the relationship between cognitive load and user engagement, which can be considered a type of information-learning outcome.

3.1.3. Expectation confirmation theory

Expectation confirmation theory, proposed by Oliver [38], emphasizes how the confirmation between actual performance and expected performance influences user satisfaction [22]. As claimed in expectation

confirmation theory, two conceptual blocks are included: initial beliefs and actual performance. When perceived performance exceeds expectations, users' satisfaction is evoked [62]. Conversely, when perceived performance fails to meet expectations, dissatisfaction occurs [12]. The expectation confirmation theory fits our research for the following two reasons. On the one hand, this theory provides us with a basis to understand the influence of cognitive load variance induced by the shift of information processing targets. When users process different information within an information set, the first appearance of information can make users generate expectations. With these expectations, users compare subsequent information with them and generate confirmation perceptions. On the other hand, expectation confirmation theory has been widely used in online review contexts to explain how users react to different information within the same information set [67], which supports our discussion on the moderating effect of cognitive load variance.

3.2. Hypothesis development

3.2.1. Cognitive load and user engagement

Intrinsic cognitive load is commonly induced by information content itself [24], such as information complexity [13] and element interactivity [50]. Sweller [50], for example, demonstrated that with the increment in information complexity, individuals need to allocate more cognitive effort. The increasing complexity of review content is expected to consume many cognitive resources [51], inducing high intrinsic cognitive load. Related to the current study, the increasing attribute-related information embedded in online reviews, considered as one information content characteristic, needs consumers to devote many cognitive resources to digest it. In this process, consumers are likely to experience high intrinsic cognitive load. Because of the limited information processing resources [2], the increasing intrinsic cognitive load enhances the cost of information processing, which inhibits users engage with the given content [7,69]. Therefore, we propose the following hypothesis (Fig. 1):

Hypothesis 1. Intrinsic cognitive load is negatively related to user engagement.

Extraneous cognitive load concerns the organization and presentation of information [51]. The well-displayed information can substantially decrease extraneous cognitive load [50]. Thees, Kapp, Strzys, Beil, Lukowicz and Kuhn [54], for example, provided direct evidence for the relationship between the presentation of information and extraneous cognitive load, indicating that information with vivid presentation (low extraneous load) can help individuals better comprehend the

information. Considering information processing load as a cost of learning novel information [7,58], it is expected that with the increment in the cost, users have a low possibility of conducting relative behaviors [7,69]. When individuals process less- or badly-presented information, they need to spend more extraneous cognitive effort (vs. well-organized information) [24]. The increasing extraneous cognitive load may exhaust available cognitive resources [13], which inhibits user engagement. Hence, we propose the following hypothesis (Fig. 1):

Hypothesis 2. Extraneous cognitive load is negatively related to user engagement.

Germane cognitive load, defined as the effort the individuals spend on processing and memorizing the information to build schemas [24], has been acknowledged to significantly affect working memory [39]. Germane cognitive load is expected to be as high as possible because it can generate and store novel information in long-term memory [13,52]. With the schema construction, automated deliberation guidance and information processing strategies would be stored in the long-term memory [13,39]. When users experience the information-inducing low germane cognitive load, they need to devote more cognitive resources to subsequent information processing tasks [24,26], which can inhibit user engagement [7,69]. Hence, we propose the following hypothesis (Fig. 1):

Hypothesis 3. Germane cognitive load is positively related to user engagement.

3.2.2. Moderating effect of the cognitive load variance

In addition to the cognitive load induced by separate information, it is also important to understand how the corresponding cognitive load changes from processing one to another information. Expectation confirmation theory [22,38] can provide us with a theoretical basis to explain the moderating effect of corresponding cognitive load variance. Related to our research, when users experience intrinsic, extraneous, and germane cognitive load induced by information processing, they may generate the expectations that the cost of subsequent information processing (i.e., cognitive resources used to digest information) would be at a similar level. If cognitive resources used for subsequent information processing exceed expected levels, individuals have a high possibility of being dissatisfied. By contrast, users may generate positive reactions when information digestion effort are below individuals' expectations.

For intrinsic cognitive load [42,47]. It means that a certain level of information complexity can help individuals build intrinsic cognitive load expectations. When the subsequent information complexity is beyond

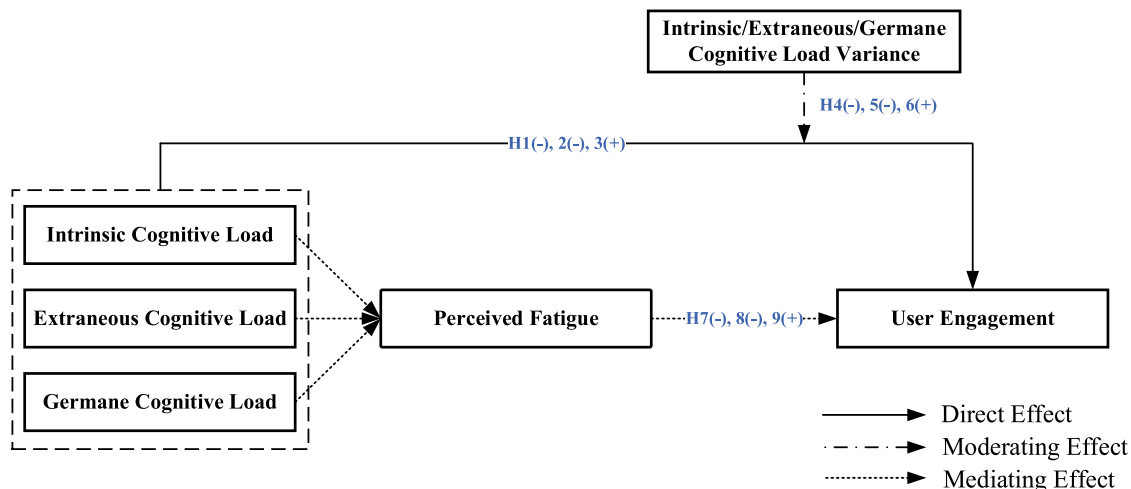


Fig. 1. Research model.

expected levels, users can generate the perception that they need to devote more intrinsic cognitive resources to subsequent information digestion. The increasing subsequent intrinsic cognitive load can induce some undesirable outcomes [24,37], which strengthens the negative influence of intrinsic cognitive load on user engagement. Hence, we propose the following hypothesis (Fig. 1):

Hypothesis 4. Intrinsic cognitive load variance accentuates the negative effect of intrinsic cognitive load on user engagement.

For extraneous cognitive load, information presentation is considered a driver [13]. As claimed in cognitive load theory [13,37], vividly presented information can decrease extraneous cognitive load. In online review contexts, when consumers initially come across well-presented online reviews, such as pictorial, animated, and distinctive information [3,45], they are immersed in the environment in which information is well-organized and low extraneous cognitive load. The perceptions formed in this environment contribute to individuals forming the expectations that the extraneous cognitive load induced by subsequent information is also at similar levels. However, when subsequent information digestion induces higher subsequent extraneous cognitive load, the disconfirmation between expectation and reality can elicit individuals' negative feelings, which accentuates the dark side of extraneous cognitive load. Hence, we propose the following hypothesis (Fig. 1):

Hypothesis 5. Extraneous cognitive load variance accentuates the negative effect of extraneous cognitive load on user engagement.

Germane cognitive load is highly related to the similarity between existing schemas and novel information [24,37]. When the newly acquired information is similar to cognitive schemas, individuals can extract the content related to novel information from existing schemas. It means that users do not need to allocate cognitive resources to process overlapping sections between novel information and schemas. Hence, when individuals encounter information related to schemas, their germane cognitive load decreases, which contributes to users generating the perception that subsequent information has a high possibility of making users consistently experience low germane cognitive load. By contrast, when reality positively disconfirms from generated expectations, some positive reactions are evoked because germane cognitive load is desirable [13]. Therefore, we propose the following hypothesis (Fig. 1):

Hypothesis 6. Germane cognitive load variance accentuates the positive effect of germane cognitive load on user engagement.

3.2.3. Effect pathway of perceived fatigue

As claimed in the stressor-strain-outcome framework [31,71], the stressors induced by load-related factors can influence individuals' perceptions, which further affect user behavior. We argue that three types of cognitive load can influence user engagement through users' perceptions of fatigue. The existing studies have widely acknowledged that load-related factors can make users experience fatigue [17]. For example, Maier, Laumer, Eckhardt and Weitzel [36] documented that users' perceptions of social load increase their feelings of social network site exhaustion. Zhang, Zhao, Lu and Yang [71] demonstrated that social overload is positively related to social network fatigue. Combined with cognitive load theory, we suggest the effect pathways of intrinsic, extraneous, and germane cognitive load should be heterogeneous.

For intrinsic cognitive load, information complexity increases intrinsic cognitive load [24,37]. Based on the premise that cognitive load is composed of intrinsic, extraneous, and germane cognitive load

[40], the increment in intrinsic cognitive load consumes available cognitive resources, which may induce cognitive resources exhaustion. In this situation, users would form the perceptions of fatigue, which can inhibit user engagement [8]. Hence, we propose the following hypothesis (Fig. 1):

Hypothesis 7. Intrinsic cognitive load increases users' perceptions of fatigue, which, in turn, inhibits user engagement.

When the focus shifts to information presentation, badly presented information makes individuals spend more cognitive resources to process information than vividly presented content, which can induce the increment in extraneous cognitive load [13,39]. When users are in the high extraneous cognitive load condition, they are likely to experience fatigue because of individuals' limited information processing ability [58,71]. Given the dark side of strain (represented by fatigue) on user engagement [9,14,71], we propose the following hypothesis (Fig. 1):

Hypothesis 8. Extraneous cognitive load increases users' perceptions of fatigue, which, in turn, inhibits user engagement.

On the contrary, because germane cognitive load is beneficial for sparing cognitive resources to further information digestion [24,37], we argue that the increasing germane cognitive load not only not consumes available cognitive resources but also contributes to further information processing. In this regard, when individuals are under the high germane cognitive load condition, they have a low possibility of generating the perceptions of fatigue. From the perspective of affect-as-information [46,65], individuals would make decisions based on their affect. Combined with the confirmed influence of fatigue on user engagement [4,71], we propose the following hypotheses (Fig. 1):

Hypothesis 9. Germane cognitive load decreases users' perceptions of fatigue, which, in turn, enhances user engagement.

4. Analysis

4.1. Study 1

JD.com provides us with an ideal context to investigate how users respond to cognitive load. JD.com is a mainstream B2C platform in China and provides consumers with high-quality products, fast delivery service, and instant customer service. Its business covers product sales, logistics, and finance services. The review data provided by JD.com has been widely used in online review research [49,58]. Firstly, the information overload phenomenon in online review forums [63] contributes to examining the effects of three types of cognitive load. Secondly, as a B2C platform, one significant feature of JD.com is self-employed, which could help us alleviate the interference effects of users' cognitive bias towards different sellers. Finally, JD.com allows users to conduct user engagement (e.g., vote for reviews and provide comments).

4.1.1. Data and variables

In the data collection section, we used a web crawler to collect the available data of two major product types (i.e., search and experience products) on JD.com. The web crawler is a Python-based technology. This technique simulates the process of extracting information from reviews by users. By selecting targeted information, the web crawler automatically extracts related information from review websites. To alleviate the concern that product types may bias our results, the present study collected the review data of two mainstream products: search and experience products [49,58,68]. Consistent with the previous studies, we choose mobile phones, televisions, and laptops to identify search

products and use skincare, rice cookers, and sports shoes to identify experience products [58]. Finally, 13,152 product reviews are obtained. In the data processing section, the invalid data, incomplete data, and repetitive data were deleted at the first step. After this step, 9691 online reviews remained.

The dependent variable is user engagement (i.e., *EngaLikes* and *EngaComments*), measured by the number of like-clicking and reply counts [25]. The main independent variables are intrinsic (i.e., *Intrinsic*), extraneous (i.e., *Extraneous*), and germane (i.e., *Germane*) cognitive load. For intrinsic cognitive load, we used the ratio of attribute count to word count to identify it as suggested by Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13]. We calculated the attribute count by installing the attribute dictionary used by Wang, Ngai and Li [58], in which all attributes are collected through user manuals, Chinese dictionaries, and the Internet. When the attributes embedded in review content were matched with the attribute dictionary, we counted them. Otherwise, the attribute count was zero. Specifically, we first built an attribute dictionary.² Then, using a funnel calculating model, we calculated attribute count. To vividly explain how attribute count is calculated, we take the following example. A reviewer claimed that “*Not bad, but the system of the TV series is a bit of garbage, the software can not be downloaded*”. In this review, two attributes are mentioned: *system* and *software*. Then, by matching these two attributes with the attribute dictionary we developed, we calculated the attribute count. If these two attributes are included in the attribute dictionary, the attribute count is two. Because information quantity can be used to identify information complexity [13], we argue that the increment in the ratio between attribute count and word count increases the difficulty of consumers selecting attribute information from given information. Hence, information complexity increases. In this situation, users need to allocate many cognitive resources to digest them, strengthening users’ perceptions of intrinsic cognitive load.

For extraneous cognitive load, we used information presentation patterns to identify extraneous cognitive load [72]. As claimed in cognitive load theory [51], vividly presented information can make individuals experience lower extraneous cognitive load than poorly presented information. In online review contexts, when images are provided, consumers can efficiently allocate their attention to targeted information, which lowers extraneous load. On the contrary, when only textual information is provided, users need to finish digesting all information to confirm whether a textual review includes targeted information. If online reviews contain text and pictures, we argue that extraneous cognitive load is low. Otherwise, extraneous cognitive load is high when only text is available.

For germane cognitive load, we used text similarity calculated by the TextSimilarity function of Tencent Cloud. The information provided by firms (firm-generated content, thereafter) in the “About the Product” section could help users learn about products [60], contributing to the construction of cognitive schemas. Based on the positive influences of schemas on information processing, individuals would experience low cognitive load when the newly-coming information is similar to existing schemas [24,39]. Hence, we suggest that germane cognitive load should be low when users learn consumer reviews that are similar to firm-generated content.

The moderating variable is intrinsic/extraneous/germane cognitive load variance (i.e., *IntrinsicVar*, *ExtraneousVar*, and *GermaneVar*). We identify them by three binary variables. For intrinsic and germane cognitive load variance, if the corresponding load induced by the latter information is higher than the former information, we calculated it as zero. Otherwise, its value is one. For example, the ratio between attribute count and word count of the first-appear review is 0.5, if the ratio of the subsequent review is 0.2, *IntrinsicVar* equals 1. Otherwise, *IntrinsicVar* equals 0. For germane cognitive load, if the germane cognitive load

induced by subsequent information is higher than the former review, *GermaneVar* equals 0. Otherwise, *GermaneVar* equals 1. For extraneous cognitive load, to avoid potential multicollinearity, we used $-1, 0$, and 1 to portray extraneous cognitive load variance. For example, if extraneous cognitive load induced by subsequent reviews is higher/lower than the former one, *ExtraneousVar* equals $1/-1$. Otherwise, *ExtraneousVar* equals 0.

Following previous research [58], we considered reviewer avatar (i.e., *Avatar*), reviewer anonymity (i.e., *Anonymity*), membership (i.e., *Membership*), review elapsed time (i.e., *Timeline*), review rating (i.e., *Rating*), rating inconsistency (i.e., *Inconsistency*), positive emotion (i.e., *PosEmo*), negative emotion (i.e., *NegEmo*), and product types (i.e., *Types*) as control variables. For avatars, review forums allow users to customize avatars by uploading pictures, which has been proven to significantly influence user engagement behavior [27]. Platforms also allow users to determine whether to display their full name. Anonymity has been proven as a detrimental factor influencing user engagement [1]. Membership identifies whether a reviewer is a Plus membership of JD.com, which can have an important effect on user engagement [58]. Timeline denotes the elapsed time of reviews, which can induce a positive effect on user engagement because of the accumulation effect [29]. Rating is a major review factor influencing user engagement [67]. Rating inconsistency depicts the difference between the review rating and the average rating, which has been proven to significantly affect user engagement [58]. Positive and negative emotion describes the emotional information embedded in online reviews, which can influence user engagement [59]. Product types refer to the category of targeted products (i.e., search or experience products). Details are shown in Table 1.

4.1.2. Analysis and results

To examine the effects of the three types of cognitive load on user engagement, we set four models to test Hypotheses 1, 2, and 3. Models 1 and 2 are the benchmark models, only containing control variables (i.e., *Controls*). In Models 3 and 4, we added intrinsic cognitive load (i.e., *Intrinsic*), and the coefficient of intrinsic cognitive load (i.e., α_1) is calculated for Hypothesis 1. In Models 5 and 6, we added extraneous cognitive load (i.e., *Extraneous*) to the basic model. And the estimation of α_2 is used for Hypothesis 2. In Models 7 and 8, we added germane cognitive load (i.e., *Germane*) to the benchmark model. The coefficient α_3 is used for Hypothesis 3. Models 9 and 10 include all three types of cognitive load, shown in Eq. (1). We follow prior research [66] using the negative binomial regression to evaluate the effects of three types of cognitive load.

$$\begin{aligned} \text{EngaLikes/EngaComments} = & \alpha_0 + \alpha_1 \times \text{Intrinsic} + \alpha_2 \times \text{Extraneous} + \alpha_3 \\ & \times \text{Germane} + \alpha \times \text{Controls} + \varepsilon \end{aligned} \quad (1)$$

Furthermore, we also discuss the moderating effect of corresponding cognitive load variance on the aforementioned relationships, as shown in Eq. (2). In particular, we used *Intrinsic* $\times$ *IntrinsicVar*, *Extraneous* $\times$ *ExtraneousVar*, and *Germane* $\times$ *GermaneVar* to present these moderating effects.

$$\begin{aligned} \text{EngaLikes/EngaComments} = & \varphi_0 + \varphi_1 \times \text{Intrinsic} + \varphi_2 \times \text{Extraneous} + \varphi_3 \\ & \times \text{Germane} + \varphi_4 \times \text{IntrinsicVar} + \varphi_5 \\ & \times \text{ExtraneousVar} + \varphi_6 \times \text{GermaneVar} + \varphi_7 \\ & \times \text{Intrinsic} \times \text{IntrinsicVar} + \varphi_8 \times \text{Extraneous} \\ & \times \text{ExtraneousVar} + \varphi_9 \times \text{Germane} \\ & \times \text{GermaneVar} + \varphi \times \text{Controls} + \varepsilon \end{aligned} \quad (2)$$

As shown in Table 2, empirical results indicate that intrinsic cognitive load negatively influences the number of likes and comments ($\alpha_{1_Likes} = -6.577, p = 0.000; \alpha_{1_Comments} = -6.640, p = 0.000$). For

² Attribute dictionary could be provided upon request.

Table 1
Descriptions of variables.

Types	Variables	Measurements	Mean	S.D.	Min	Max
Dependent	<i>EngaLikes</i>	The number of like-clicking.	2.106	28.915	0.000	1504
	<i>EngaComments</i>	The number of review replies.	1.046	14.715	0.000	739
Independent	<i>Intrinsic</i>	The number of attributes embedded in the review text divided by review length	0.076	0.067	0.000	1.750
	<i>Extraneous</i>	The organization and presentation of information (text = 1; text+image = 0).	0.606	0.489	0.000	1.000
	<i>Germane</i>	The similarity between firm-generated content and user-generated content (reverse-coded).	0.546	0.048	0.419	0.903
Moderating	<i>IntrinsicVar</i>	The intrinsic cognitive load changes from processing one to another information (increasing = 0, decreasing = 1).	0.539	0.499	0.000	1.000
	<i>ExtraneousVar</i>	The extraneous cognitive load changes from processing one to another information (increasing = 1, decreasing = -1, and equal = 0).	-0.767	0.642	-1.000	1.000
	<i>GermaneVar</i>	The germane cognitive load changes from processing one to another information (increasing = 0, decreasing = 1).	0.525	0.499	0.000	1.000
Control	<i>Avatar</i>	Whether reviewers upload personalized avatars (not upload = 0; upload = 1).	0.462	0.499	0.000	1.000
	<i>Anonymity</i>	Whether reviewers are anonymous (not anonymous = 0, anonymous = 1).	0.949	0.221	0.000	1.000
	<i>Membership</i>	Whether reviewers are JD.com members (not JD.com members = 0, JD.com members = 1).	0.401	0.490	0.000	1.000
	<i>Timeline</i>	The difference between review collection time and review posting time.	4.011	1.223	0.000	6.943
	<i>Rating</i>	Review rating from one-star to five-star.	4.697	0.950	1.000	5.000
	<i>PosEmo</i>	The positive emotion embedded in online reviews.	0.199	0.284	0.000	0.999
	<i>NegEmo</i>	The negative emotion embedded in online reviews.	0.801	0.284	0.000	1.000
	<i>Types</i>	The types of targeted products (search products = 1, experience products = 0).	0.556	0.497	0.000	1.000

extraneous cognitive load, its effect on the number of likes and comments is also negative ($\alpha_{2_Likes} = -2.159, p = 0.000$; $\alpha_{2_Comments} = -1.885, p = 0.000$). In contrast, germane cognitive load positively affects the number of likes and comments ($\alpha_{3_Likes} = 4.945, p = 0.000$; $\alpha_{2_Comments} = 2.105, p = 0.006$). When taking all dependent variables together, the examined relationships also exist. Hence, Hypotheses 1, 2, and 3 are supported.

As shown in Table 3, the results indicate that intrinsic cognitive load variance accentuates the negative effect of intrinsic cognitive load on user engagement ($\varphi_{7_Likes} = 18.523, p = 0.000$; $\varphi_{7_Comments} = 20.481, p = 0.000$). For extraneous cognitive load, extraneous cognitive load variance strengthens the negative effect of extraneous cognitive load ($\varphi_{8_Likes} = -1.082, p = 0.000$; $\varphi_{8_Comments} = -1.058, p = 0.000$). Results also indicate that germane cognitive load variance enhances the positive effect of germane cognitive load ($\varphi_{9_Likes} = -3.779, p = 0.059$; $\varphi_{8_Comments} = -6.016, p = 0.001$). Including all independent variables, the moderating effects still hold. Therefore, Hypotheses 4, 5, and 6 are supported.

4.1.3. Alternative measurements

To ensure the robustness of our main findings, we employed an alternative measurement of intrinsic, extraneous, and germane cognitive load. In detail, we used the word count of reviews containing product or platform attributes following Wang, Ngai and Li [58] to identify intrinsic cognitive load. When consumers search for targeted information from online reviews, attribute-related information is considered an important information source [35]. We argue that with the increment in the word count of reviews containing attributes, information complexity increases, which makes consumers experience high intrinsic cognitive load. For extraneous cognitive load, we used the variance of picture count to depict extraneous cognitive load. With the variance increases, the reviews with high variance have a higher possibility of attracting user attention than low-variance reviews. This follows the same effect logic as claimed in the von Restorff isolation effect [6], that is, when focal information is distinct from the background information, focal information has a high possibility to stand out and attract user attention. In this situation, consumers would experience low extraneous cognitive load. For germane cognitive load, it depicts how novel information confirms the existing schemas [13]. Based on the premise that user experience depicts consumers' knowledge of product quality evaluation, product information seeking, and online review processing, we expect that experienced consumers have a higher possibility to have schemas similar to the newly coming information, which makes consumers experience low germane cognitive load. Consistent with the main analysis, we used the number of likes and reply count to

identify user engagement and employed the negative binomial regression model to analyze the data set. Results prove the robustness of our main findings, indicating that intrinsic and extraneous (germane) cognitive load negatively (positively) influence user engagement,³ see Table 4.

4.2. Study 2

Study 2 primarily addresses measurement bias and reveals the underlying mechanisms through which different types of cognitive load influence user engagement, as proposed in Hypotheses 7–9. To do so, we varied the three types of cognitive load to elicit participants' heterogeneous perceptions of fatigue in three between-subjects designs.

4.2.1. Main study design and procedure

To ensure that participants across conditions received the same stimuli except for the cognitive load in each situation, we delicately designed the experiment materials based on thirty-two reviews used by Liu and Karahanna [35]. Before we generated the stimulus materials, we only kept positive reviews to alleviate the dissonance of online reviews [66], leaving twenty-one available online reviews. To increase the possibility of participants reading all provided online reviews, we randomly selected eight reviews from twenty-one reviews.

Then, we generated related materials to manipulate cognitive load. More specifically, we altered the number of online reviews participants need to read following Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13] to vary intrinsic cognitive load. In detail, in the low intrinsic cognitive load condition, participants were asked to read one online review. By contrast, they were asked to read all eight online reviews in the high intrinsic cognitive load condition. In this process, we kept all others the same.

For extraneous cognitive load, we, based on Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13], changed the organization of these eight online reviews by "summarization function" (simply list reviews vs. reviews organized based on product attribute). This function is expected to help participants efficiently find information matching

³ We used picture count variance to identify extraneous cognitive load. When count variance is high, it is easier for consumers to find needed information, which decreases extraneous cognitive load. Hence, the coefficient of the effect of extraneous cognitive load should be positive. Consistency, we used experience value to depict germane cognitive load. In detail, with the increment in experience value, germane cognitive load decreases, which inhibits user engagement. Hence, the coefficient should be negative.

Table 2
Effects of Three types of cognitive load.

	Model (1)		Model (2)		Model (3)		Model (4)		Model (5)		Model (6)		Model (7)		Model (8)		Model (9)		Model (10)	
	EngaLikes		EngaComments		EngaLikes		EngaComments		EngaLikes		EngaComments		EngaLikes		EngaComments		EngaLikes		EngaComments	
<i>Intrinsic</i>																				
<i>Extraneous</i>																				
<i>Germane</i>																				
<i>Avatar</i>	1.030***	0.898***			0.931***	0.823***			0.606***	0.549***			4.945***	2.105***			2.493***	2.105***	1.985***	
<i>Anonymity</i>	-0.419**	-0.447***			-0.495***	-0.539***			-0.274	-0.117			1.110***	0.894***			-10.07***	9.696***	-9.056***	
<i>Membership</i>	0.0173	-0.0197			0.0112	-0.0549			-0.295***	-0.0713			-0.376**	-0.423**			0.671***	0.508***	0.508***	
<i>Unlabeled</i>	0.974***	0.953***			0.984***	0.952***			0.851***	0.845***			0.0363	-0.000777			-0.243***	-0.243***	-0.0963	
<i>Rating</i>	-0.450***	-0.529***			-0.448***	-0.534***			-0.564***	-0.690***			0.940***	0.902***			0.856***	0.811***	0.811***	
<i>Negemo</i>	4.760	4.815			4.450	4.429			2.231	3.210			14.77	18.75			12.22	15.95	15.95	
<i>Posemo</i>	4.974	4.616			4.714	4.239			2.371	3.241			15.04	18.58			12.41	15.92	15.92	
<i>Types</i>	1.762***	1.541***			1.634***	1.401***			1.497***	1.294***			1.989***	1.653***			1.501***	1.168***	1.168***	
<i>_Cons</i>	-8.286	-7.889			-7.441	-6.836			-3.461	-4.451			-21.05	-22.83			-17.73	-18.89	-18.89	
<i>lnalpha</i>	2.682***	2.441***			2.651***	2.404***			2.448***	2.225***			2.640***	2.410***			2.341***	2.150***	2.150***	
<i>N</i>	9666	9645			9645	9645			9666	9645			9096	9075			9075	9075	9075	
<i>pseudo R²</i>	0.068	0.076			0.071	0.080			0.100	0.108			0.072	0.078			0.117	0.118	0.118	

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

participants' needs without changing the information content itself. To create the function, we recruited three research assistants to select the most important features embedded in review content. Three research assistants were asked to write down the most four important features independently. When conflict opinions occurred, the three assistants were allowed to discuss with each other to make the final decisions. Then, we pooled the top four most important key points and made the summarization linking the key attribute with corresponding online reviews. For example, for the attribute: battery, we linked the keyword with the reviews containing the battery description. In the high extraneous cognitive load condition, participants were asked to read these eight reviews without this summarization function. On the contrary, in the low extraneous load condition, this summarization function was provided to help participants match the key points with detailed information.

For germane cognitive load, we manipulated it with or without procedural guidance as suggested by Cheng, Fu, de Vreede, de Vreede, Seeber, Maier and Weber [13]. In the high germane cognitive load condition, participants were instructed to learn product information with the tip—that is, “Please read all the reviews below carefully. You can focus on camera attributes such as photo quality, portability, standby time, cost-efficiency, and ease of use”. By contrast, in the low germane cognitive load condition, participants are provided with the tip, that is, “Please read all the reviews below carefully”.

Three hundred and fifty-three participants (62.15 % female) were recruited from Credamo. Participants were randomly assigned to the low or high condition of the three types of cognitive load. Participants were asked to read online reviews and to evaluate their perceptions of intrinsic, extraneous, and germane cognitive load using three (i.e., “The content of online reviews was very complex”, “The content of online reviews covered formulas that I perceived as very complex”, and “The content of online reviews covered concepts and definitions that I perceived as very complex”.) [34], four (i.e., “I do not like the presentation or organization of these online reviews”, “I am not interested in the presentation or organization of these online reviews”, “The presentation or organization of these online reviews requires strenuous efforts when reading”, and “The presentation and organization of these online reviews are not helpful to me”.) [11], and six (i.e., “The guidance contributed to my processing of these online reviews”, “The guidance helped me to mentally organize the review processing”, “The guidance really enhanced my deliberation of these online reviews”, “As we progressed throughout the activities, the guidance helped me to concentrate on the processing of these online reviews”, “While proceeding throughout the activities, the guidance helped me to better complete my task of processing these online reviews”, and “The guidance makes it easy for me to do similar tasks again”.) [13] measurement items, respectively. Participants were also asked to rate the level of perceived fatigue (i.e., “Extremely not tired/Extremely tired”, “Extremely not bored/Extremely bored”, “Extremely not alert/Extremely alert”, and “Extremely not energetic/Extremely energetic”) [28] and their willingness to engage with the given information (i.e., “How likely would you be to engage with the reviews you see, such as liking, forwarding, or commenting on it/them?”) [61]. Besides, several important control variables were also considered, such as shopping experience (i.e., *Experience*), shopping frequency (i.e., *Familiarity*), review helpfulness (i.e., *Helpfulness*), and emotion evoked by reading reviews (i.e., *Emotion*). All these questions were measured based on a seven-point measurement scale. Moreover, participants were asked to answer two attention-check questions about the content of online reviews (i.e., “Are all reviews positive?”) and to make a summation of key camera features and review details to increase the possibility of participants reading all listed reviews. Demographic questions (i.e., *Gender*, *Education*, *Job*, and *Age*) were also asked.

4.2.2. Results

We used two hundred and fifty-seven subjects who passed both attention checks in data analysis. Results for the manipulation check of the low and high intrinsic ($M_{\text{Intrinsic_Low}} = 2.352$ versus $M_{\text{Intrinsic_High}} =$

Table 3

Moderating effects of corresponding cognitive load variance.

	Model (11)	Model (12)	Model (13)	Model (14)	Model (15)	Model (16)	Model (17)	Model (18)
	EngaLikes	EngaComments	EngaLikes	EngaComments	EngaLikes	EngaComments	EngaLikes	EngaComments
<i>Intrinsic</i> × <i>IntrinsicVar</i>	18.523***	20.481***					14.49***	15.13***
<i>Extraneous</i> × <i>ExtraneousVar</i>			−1.082***	−1.058***			−0.645***	−0.657***
<i>Germane</i> × <i>GermaneVar</i>					−3.779*	−6.016***	−14.99***	−5.516**
<i>Intrinsic</i>	−15.10***	−16.11***					−17.28***	−17.08***
<i>IntrinsicVar</i>	−1.922***	−2.048***					−1.446***	−1.588***
<i>Extraneous</i>			−2.567***	−2.078***			−2.421***	−1.773***
<i>ExtraneousVar</i>			0.0564	0.189**			0.0539	0.0689
<i>Germane</i>					6.948***	4.352***	10.87***	8.257***
<i>GermaneVar</i>					2.141**	3.180***	7.931***	3.018**
<i>Avatar</i>	0.797***	0.682***	0.392***	0.467***	1.084***	0.871***	0.447***	0.559***
<i>Anonymity</i>	−0.781***	−0.788***	−0.462**	−0.277	−0.384**	−0.413**	−0.644***	−0.451**
<i>Membership</i>	−0.132	−0.207**	−0.125	−0.0793	0.0762	0.0397	−0.101	−0.198*
<i>LnTimeline</i>	0.945***	0.931***	1.483***	1.568***	0.939***	0.901***	1.319***	1.435***
<i>Rating</i>	−0.448***	−0.534***	−0.559***	−0.625***	−0.467***	−0.532***	−0.617***	−0.642***
<i>NegEmo</i>	3.881	3.944	2.241	4.194	14.33	15.70	5.665	17.72
<i>PosEmo</i>	3.926	3.647	2.864	4.079	14.49	15.45	6.074	17.41
<i>Types</i>	1.467***	1.263***	3.942***	2.019***	1.965***	1.618***	3.365***	1.405***
<i>_Cons</i>	−4.937	−4.554	−9.469	−9.920	−21.65	−20.95	−15.38	−24.90
<i>Lnalpha</i>	2.561***	2.271***	2.346***	2.219***	2.637***	2.400***	2.214***	2.099***
<i>N</i>	9645	9645	5729	5723	9096	9075	5169	5169
<i>Pseudo R²</i>	0.080	0.094	0.101	0.104	0.073	0.079	0.115	0.113

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.**Table 4**

Results of robustness analysis.

	Model (19)	Model (20)	Model (21)	Model (22)	Model (23)	Model (24)
	EngaLikes	EngaComments	EngaLikes	EngaComments	EngaLikes	EngaComments
<i>Intrinsic</i>	−0.00590**	−0.00689**				
<i>Extraneous</i>			0.237***	0.243***		
<i>Germane</i>					−0.189***	−0.221***
<i>Avatar</i>	0.544***	0.632***	0.510***	0.599***	0.595***	0.688***
<i>Anonymity</i>	−0.525***	−0.628***	−0.312**	−0.321**	−0.540***	−0.562***
<i>Membership</i>	0.176**	0.144*	0.179**	0.121	0.342***	0.336***
<i>Length</i>	0.0259***	0.0254***	0.0175***	0.0178***	0.0229***	0.0218***
<i>Lntimeline</i>	0.843***	0.859***	0.775***	0.726***	0.843***	0.878***
<i>Rating</i>	−0.485***	−0.571***	−0.541***	−0.669***	−0.482***	−0.560***
<i>Negemo</i>	3.642	4.706	2.375	3.208	3.797	4.884
<i>Posemo</i>	3.727	4.062	2.411	2.593	3.862	4.225
<i>Types</i>	1.260***	0.886***	1.053***	0.523***	1.389***	1.007***
<i>_Cons</i>	−7.192	−7.537	−5.887	−5.637	−5.650	−5.895
<i>Lnalpha</i>	2.261***	1.965***	2.074***	1.750***	2.257***	1.955***
<i>N</i>	9645	9645	9666	9645	9665	9644
<i>pseudo R²</i>	0.120	0.143	0.145	0.177	0.121	0.144

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.271; $SD_{\text{Intrinsic_Low}} = 1.257$ versus $SD_{\text{Intrinsic_High}} = 1.497$; $t = -5.686$; $p = 0.000$)/extraneous ($M_{\text{Extraneous_Low}} = 2.231$ versus $M_{\text{Extraneous_High}} = 3.169$; $SD_{\text{Extraneous_Low}} = 0.598$ versus $SD_{\text{Extraneous_High}} = 1.333$; $t = -3.645$; $p = 0.001$)/germane cognitive load ($M_{\text{Germane_Low}} = 5.899$ versus $M_{\text{Germane_High}} = 6.226$; $SD_{\text{Germane_Low}} = 0.745$ versus $SD_{\text{Germane_High}} = 0.411$; $t = -2.773$; $p = 0.007$) indicated that manipulations are successful.

Moreover, we also ensure the reliabilities, convergence, and discriminant validities of the focal variables. First, we calculate the Cronbach's alphas. The alphas of the intrinsic cognitive load and perceived fatigue were 0.937 and 0.887, respectively. In the extraneous cognitive load condition, Cronbach's alphas of extraneous cognitive load (perceived fatigue) were 0.866 (0.866). In the germane cognitive load condition, Cronbach's alphas of germane cognitive load and perceived fatigue were 0.808 and 0.763, respectively. All values of Cronbach's alphas were higher than 0.7 [66], ensuring high reliabilities. Second, we calculated the composite reliability (CR) and average variance extracted scores (AVE) to ensure convergent validity. In the intrinsic cognitive load, the intrinsic cognitive load's (perceived fatigue's) CR and AVE were 0.931 (0.890) and 0.820 (0.670), respectively.

In the extraneous cognitive load condition, the CR and AVE of the extraneous cognitive load (perceived fatigue) were 0.849 (0.847) and 0.585 (0.583). In the germane cognitive load condition, the CR and AVE of the germane cognitive load (perceived fatigue) were 0.860 (0.850) and 0.508 (0.588). Because all AVEs were higher than 0.5 [20] and CRs were higher than 0.8 [68], we ensured convergent validity. Moreover, as shown in Table 5, all the factor loadings were higher on corresponding constructs than on others, confirming the convergence validities of these variables [68]. Third, we also evaluated the discriminant validity. In the intrinsic cognitive load condition, the square roots of the AVE for the intrinsic cognitive load (perceived fatigue) were 0.906 (0.819), higher than the correlations between them (i.e., 0.571). In the extraneous cognitive load condition, the square roots of the AVE for extraneous cognitive load and perceived fatigue were 0.765 and 0.764, respectively. All the them are higher than the magnitude of the correlations between them (i.e., 0.757). In the germane cognitive load condition, the square roots of the AVEs for the germane cognitive load and perceived fatigue were 0.713 and 0.766, respectively, which were greater than the magnitude of the correlations between them (i.e., 0.177). All the results confirmed the discriminant validity.

Table 5
Exploratory Factor Analysis Results.

Construct	Item #	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
Intrinsic (Cronhach = 0.937, CR = 0.931, AVE = 0.820)	1	0.827	0.438	–	–	–	–
	2	0.962	0.122	–	–	–	–
	3	0.922	0.217	–	–	–	–
Fatigue_Intr (Cronhach = 0.887, CR = 0.890, AVE = 0.670)	1	0.520	0.733	–	–	–	–
	2	0.091	0.841	–	–	–	–
	3	0.340	0.789	–	–	–	–
	4	0.179	0.901	–	–	–	–
Extraneous (Cronhach = 0.866, CR = 0.849, AVE = 0.585)	1	–	–	0.737	0.524	–	–
	2	–	–	0.841	0.128	–	–
	3	–	–	0.695	0.499	–	–
	4	–	–	0.780	0.297	–	–
Fatigue_Extr (Cronhach = 0.866, CR = 0.847, AVE = 0.583)	1	–	–	0.523	0.765	–	–
	2	–	–	0.517	0.636	–	–
	3	–	–	0.217	0.781	–	–
	4	–	–	0.197	0.855	–	–
Germane (Cronhach = 0.808, CR = 0.860, AVE = 0.508)	1	–	–	–	–	0.825	0.027
	2	–	–	–	–	0.657	–0.149
	3	–	–	–	–	0.674	–0.022
	4	–	–	–	–	0.636	–0.085
	5	–	–	–	–	0.753	–0.083
	6	–	–	–	–	0.715	–0.065
Fatigue_Germ (Cronhach = 0.763, CR = 0.850, AVE = 0.588)	1	–	–	–	–	–0.032	0.785
	2	–	–	–	–	–0.214	0.670
	3	–	–	–	–	–0.004	0.717
	4	–	–	–	–	–0.040	0.879

Note. Fatigue_Intr, Fatigue_Extr, and Fatigue_Germ refer to the construct of perceived fatigue in the Intrinsic, extraneous, and germane cognitive load conditions.

Table 6
Descriptive of control variables ($N = 237$).

	N	Mean	Min	Max	S.D.
Helpfulness	237	5.803	2.333	7	0.788
Familiarity	237	2.05	1	6	1.185
Experience	237	2.43	1	6	1.074
Emotion	237	4.95	1	7	0.951
Gender	237	1.35	1	2	0.479
Education	237	5.08	4	6	0.462
Job	237	3.69	1	6	1.715
Age	237	3.14	1	6	1.146

Before we conducted the analysis, we, first, summarized control variables (shown in Table 6). To test Hypothesis 7, we employed ordinary least squares to analyze the collected data because the dependent variable (i.e., user engagement) is continuous. First, we check the difference of control variables between the low and high intrinsic cognitive load conditions. The results indicate that there is no significant difference in control variables ($p > 0.05$). Next, we used Stata to examine the effect of intrinsic cognitive load on user engagement, indicating that intrinsic cognitive load negatively influences user engagement ($\beta = -0.207$, $p = 0.094$). In addition, using PROCESS MODEL 4 with 1000 bootstrap, we suggest that perceived fatigue negatively mediates the effect of intrinsic cognitive load on user engagement ($\beta = -0.446$, Confidence Interval = -0.659 to -0.258). Specifically, intrinsic cognitive load increases users' perceptions of fatigue ($\beta = 0.440$, Confidence Interval = 0.284 to 0.595), which, in turn, decreases user engagement ($\beta = -1.013$, Confidence Interval = -1.308 to -0.719).

Likewise, the independent samples t -test indicates that there is no significant difference in control variables ($p > 0.05$) between the low and high extraneous cognitive load conditions except for review helpfulness and users' evoked emotion ($p < 0.05$). Analysis not only reveals that extraneous cognitive load negatively impacts user engagement ($\beta = -0.809$, $p = 0.000$) but also indicates that extraneous cognitive load accentuates users' perceptions of fatigue ($\beta = 0.580$, Confidence Interval = 0.406 to 0.753), which, in turn, attenuates user engagement ($\beta = -0.539$, Confidence Interval = -0.787 to -0.291).

Following the same logic, we, first, compare the difference of control variables between the low and high germane cognitive load conditions.

Independent samples t -test indicates that there is no significant difference in control variables ($p > 0.05$) except for user familiarity ($p < 0.05$). Analysis suggests that germane cognitive load positively but insignificantly influences user engagement ($\beta = 0.205$, $p = 0.210$). Mediation analysis reveals that perceived fatigue positively mediates the effect of germane cognitive load on user engagement ($\beta = 0.133$, Confidence Interval = 0.019 to 0.288). More specifically, germane cognitive load alleviates users' perceptions of fatigue ($\beta = -0.221$, Confidence Interval = -0.477 to 0.036),⁴ which, thereafter, promotes user engagement ($\beta = -0.601$, Confidence Interval = -0.828 to -0.374). Therefore, Hypotheses 7, 8, and 9 are validated.

4.2.3. Robustness check

To increase the validity of our results, we conducted another online experiment. Three hundred and seventy participants were invited to participate in our experiment (65 % female) from Credamo. In this new experiment, we kept all consistent with the previous experiment in addition to key measurements. In the new experiment, we used four (i.e., “It is difficult to process the above reviews”, “The above reviews are complex”, “The above reviews are not easy to understand”, and “Understanding the above reviews is beyond my capability”) [13], three (i.e., “The presentation or organization of the above reviews is very unclear”, “From an information processing perspective, the presentation or organization of the above reviews is very inefficient”, and “The presentation or organization of the above reviews is full of ambiguity”) [34], and four (i.e., “The guidance indeed enhanced my understanding of the above reviews”, “The guidance indeed enhanced my knowledge and understanding of the above reviews”, “The guidance indeed enhanced my understanding of the content of the above reviews”, and “The guidance indeed enhanced my understanding of the concepts and definitions in the above reviews”) [34] items to measure intrinsic, extraneous, and germane cognitive load. For perceived fatigue and user engagement, six (i.e., “I feel tired when I read the above reviews”, “I feel bored when I read the above reviews”, “I feel exhausted when I read the above reviews”, “I feel drained when I read the above reviews”, “I am not interested in whether the above reviews mention new information about the

⁴ Although the confidence interval does not include zero, the significance level is 0.091, indicating marginally significant.

camera”, and “I do not care whether the above reviews mention new information about the camera.”) [48] and five (i.e., “After reading the above reviews, my intention to engage with them, such as liking, sharing, and commenting”, “After reading the above reviews, my intention to discuss them”, “After reading the above reviews, my likelihood of discussing them online”, “After reading the above reviews, my likelihood of discussing them online”, and “After reading the above reviews, my willingness to continuously discuss them online.”) [70] items were used to replace items used in the previous experiment. Three hundred and forty-six subjects passed two attention checks and, thus, were used in the following analyses. Using PROCESS MODEL 4, analyses revealed that: (1) intrinsic cognitive load increases users’ perceptions of fatigue ($\beta = 0.735$, Confidence Interval = 0.600 to 0.870), which subsequently inhibits user engagement ($\beta = -0.459$, Confidence Interval = -0.740 to -0.177), (2) extraneous cognitive load increases perceived fatigue ($\beta = 0.614$, Confidence Interval = 0.489 to 0.739), which, in turn, reduces user engagement ($\beta = -0.653$, Confidence Interval = -0.867 to -0.440), and (3) with the increment in germane cognitive load, users’ perceptions of fatigue are alleviated ($\beta = -0.715$, Confidence Interval = -0.984 to -0.446), which, in turn, enhances user engagement ($\beta = -0.518$, Confidence Interval = -0.664 to -0.371). The results provide strong evidence supporting our findings.

5. Discussion

5.1. Conclusion

With the increment in available information in online review platforms, cognitive load becomes salient and concerns platforms and consumers. The present study, based on the cognitive load theory, discusses the influences of intrinsic, extraneous, and germane cognitive load on user engagement, their accentuators and attenuators, and their underlying mechanisms. In detail, analyzing the data set collected from JD.com and two online experiments, we indicated that: (1) intrinsic and extraneous cognitive load negatively influences user engagement, while germane cognitive load has a positive effect, (2) intrinsic/extraneous cognitive load variance accentuates the negative effect of intrinsic/extraneous cognitive load, while germane cognitive load variance strengthens the positive effect of germane cognitive load, and (3) perceived fatigue negatively mediates the effects of intrinsic and extraneous cognitive load but positively mediates the effect of germane cognitive load.

5.2. Theoretical implications

The present study is novel in the following aspects. Firstly, our study contributes to the user engagement literature by considering its antecedents from the cognitive load perspective. The existing studies have discussed the determinants of user engagement [25,64]. Wang, Ngai and Li [59], for example, explored the heterogeneous effects of review sentiment on online forwarding behavior. However, a few studies have discussed it from the cognitive load perspective. With the increment in the available information in online review forums, the cognitive load induced by processing the available information becomes salient [63] and the discussions on the effect of cognitive load on user engagement become important. To close the research gap, the present study, based on the cognitive load theory, enters the discussion by understanding how the three types of cognitive load influence user engagement.

Secondly, this study also enriches the cognitive load literature by considering the variance of cognitive load from processing one to another information. The existing studies, no matter distinguishing between the different types of cognitive load, have analyzed the effect of load-related factors [9,13,24,36,71]. However, most of them focus on a single piece of information, ignoring that users may refer to an information set to make decisions [66]. Based on the premise that information would interact with others within an information set, we expect that

the cognitive load variance induced by processing information from one to another can strengthen or weaken the effect of cognitive load on user engagement.

Thirdly, our study advances the load-related factors and user engagement literature by not only considering the heterogeneous effects of different types of cognitive load but also uncovering underlying mechanisms. On the one hand, most of the existing studies consider cognitive load as an overall item [2,24,58], without differentiating the different types of cognitive load. As claimed in cognitive load theory, the influences of the different types of cognitive load should be heterogeneous [13]. Hence, an in-depth discussion to differentiate these effects on user engagement is needed. On the other hand, related studies have uncovered the underlying mechanisms of the effects of load-related factors, such as social media load and information load [71]. In three pioneer studies, Maier, Laumer, Eckhardt and Weitzel [36], Zhang, Zhao, Lu and Yang [71], and Cao, Khan, Ali and Khan [9] have discussed the direct effects of load-related factors and their underlying mechanisms. Cao, Khan, Ali and Khan [9], for example, demonstrated that distress positively mediates the effect of social overload on social network site discontinuous usage intentions. However, perceived fatigue—considered an important outcome induced by load-related factors—is scarcely discussed. To generate a deep understanding of cognitive load, the present study, based on cognitive load theory and the stressor-strain-outcome framework, discusses the effect pathways of three types of cognitive load.

5.3. Practical implications

Our study also provides some guidance to practice. On the one hand, we suggest that user-generated content (UGC) platforms not only need to provide useful information to users but also need to pay attention to the cognitive load experienced by users. More specifically, to alleviate intrinsic cognitive load, platforms can pre-select informative reviews to decrease the load on users processing unhelpful information from millions of alternatives. For extraneous cognitive load, UGC platforms can optimize information presentations. The vivid information can help users efficiently locate needed information, beneficial for promoting user engagement. For germane cognitive load, forums can provide some guidance on teaching users how to effectively find and process targeted information.

On the other hand, our results indicate that intrinsic and extraneous cognitive load variance accentuates the negative effects of the intrinsic and extraneous cognitive load. By contrast, germane cognitive load variance accentuates the positive effect of the germane cognitive load. It suggests that UGC platforms should pay attention to reorder-provided information. For intrinsic and extraneous cognitive load, platforms can provide the information inducing moderate intrinsic and extraneous cognitive load in a forward position, followed by the information leading to the low corresponding cognitive load. For germane cognitive load, platforms can, first, provide the information inducing the low germane cognitive load, followed by the high-germane cognitive load reviews.

5.4. Limitations and future research

This study also has some limitations. First, given the limitations of the data set, we used attribute count, information presentation, and information similarity to portray intrinsic, extraneous, and germane cognitive load. Future research can propose alternative methods to identify these measurements. Second, we used questionnaires to ask participants to disclose their reactions to the stimuli. Future studies can use eye-tracking technologies to portray users’ reactions. Third, the present study mainly focuses on the effect pathway of users’ perceptions of fatigue. Future studies can advance our research by uncovering alternative effect pathways.

CRediT authorship contribution statement

Yuqiu Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis.
Kai Li: Writing – review & editing, Supervision, Resources, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors thank the editors and reviewers for their helpful comments. This work was supported by the Major Program of the National Natural Science Foundation of China (72091311).

Data availability

Data will be made available on request.

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